

Alok Kumar Mishra

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EDUCATION

- Indian Institute of Information Technology, Jabalpur
Bachelor of Technology - Electronics & Communication Engineering Madhya Pradesh, India
Aug 2024 – Present

EXPERIENCE

- DigiLocker (MeitY), Government of India Remote
Security Contributor (AppSec) Feb 2025 – Apr 2025
 - Application Security & Android RE:** Responsibly disclosed **10+ P1/P2 vulnerabilities** in DigiLocker's Android app (**500M+ users**) via dynamic/static analysis using **Frida, JEB, Radare2**.
 - Remediation & Recognition:** Coordinated with DigiLocker engineers on responsible fixes; received **official Govt. of India acknowledgment** for responsible disclosure.
- Rizin Project (Open-Source RE Framework) Remote
Contributor Nov 2025 – Apr 2026
 - Core Contributions:** Landed **26 merged PRs**: hardened ELF parsing, added De Bruijn helper, unified address-description API, introduced enum hints (**ahie**), and patched **7** memory-safety issues; ranked **top 10 contributor**.
 - Ecosystem:** Extended **rz-find** (multi-path), **rz-ar** (archive extraction), Rizin Notebook UX, and Windows console VT; migrated **rz-silhouette** IPC from Protocol Buffers to **Cap'n Proto**.
- Google Summer of Code 2026 – Rizin Remote
Contributor May 2026 – Present
 - Scope:** Building a **native Frida debugger backend** in C to unify dynamic instrumentation with Rizin's static analysis pipeline.
 - Implementation:** Engineering Frida attach/spawn/launch lifecycle, script execution, and **Android Java/Kotlin** class-loader recovery; piping structured class/type data into Rizin's analysis model and **Cutter UI**.

PROJECTS

- RzWeb – Rizin on Web (WebAssembly Reverse Engineering Platform)** *GitHub*
WebAssembly (WASM), WASI, TypeScript, Python, Shell, Emscripten
 - Built **RzWeb**, a browser-native reverse engineering IDE by compiling the **Rizin** toolchain to **WebAssembly (WASM)** (WASI/Emscripten), enabling secure static analysis and inspection directly in the web.
 - Engineered **RzWasi** execution and sandbox-safe host bindings to run Rizin modules inside a WASI runtime, solving syscall/runtime constraints and enabling reproducible builds across environments.
 - Implemented core RE workflows (disassembly, interactive console, graph navigation) with a responsive TypeScript UI, optimized for instant loading and near-instant analysis on cached smaller binaries.
 - Shipped a production demo at rizin.pages.dev; achieved **500+ stars**, **50+ forks**, and **100K+ visits** in the first week itself, reducing setup friction from hours to **under 2 minutes**.
- Arthyx – Quant Finance Document Intelligence** *GitHub*
Next.js, TypeScript, Neo4j, Vector Search
 - Engineered a **RAG-based quant system** for deep analysis of **500+ page financial PDFs** (Basel III, NPA, CRAR).
 - Modeled financial entities and risk propagation using **Neo4j knowledge graphs**, reducing manual review by **40%**.

TECHNICAL SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Bash, SQL, Smali, HTML/CSS
- Cybersecurity / RE:** VAPT, Bug Bounty, AppSec, Android RE, Frida, Rizin, Ghidra, IDA Pro, JEB, Radare2, Burp, Metasploit
- Web / Backend:** React, Next.js, Node.js, FastAPI, Firebase, Tailwind, REST APIs
- AI / Finance:** RAG Systems, NLP, Vector Search, Neo4j, Gemini API, Basel III, NPA/CRAR
- Tools:** Git, Docker, Linux, Android Studio, Postman, Vercel

ACHIEVEMENTS

- Acknowledged **Security Contributor** by DigiLocker (GOI) for 10+ critical vulnerabilities (**500M+ users**).
- 1st Place** – Can You Hack It Hackathon (200+ students), Team Lead.
- Top 20 / 700+ teams** – HackTheFuture 2025, IIT Gandhinagar.
- JEE Main Top 1.2%**; CBSE Merit (**100/100 CS**); **Global Rank 385** in CodeChef Starters 176C.

VOLUNTEER EXPERIENCE

- Jagrati – Non-Profit Education Initiative** Jabalpur, India
Mathematics Tutor (Volunteer) Aug 2024 – Present
 - STEM Education Outreach:** Providing free mathematics tutoring to underprivileged students for Navodaya entrance exams; designed simplified study plans to improve conceptual understanding.